



S/4HANA ACCELERATION SERIES · VOLUME 3

AI-Accelerated S/4HANA Migration: A Practitioner's Playbook

How to use GenAI and agentic tools to compress timelines, cut manual effort, and reduce risk across every phase of your S/4HANA program.

Prepared by labwyze

Industry-agnostic edition

Synthesis of SAP, Big 4, SI, and independent practitioner sources

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HOW TO READ THIS PLAYBOOK

This is Volume 3 of the Labwyze S/4HANA Acceleration Series. Volume 1 covers the migration decision framework (greenfield, brownfield, bluefield, RISE, GROW). Volume 2 covers AI readiness for the finance function post-go-live. This volume focuses on AI as an accelerator during the migration program itself. Figures are indicative benchmarks; actual results depend on program scope, data quality, and tooling maturity. External claims are numbered to the source list in the Appendix.

1 Executive Summary

The two largest consumers of time and money in an S/4HANA migration, custom code remediation and testing, can now be substantially automated. The technology is in production today, SAP's own tooling is shipping agentic capabilities in 2026, and every major system integrator has built or acquired an AI acceleration layer. The open question is not whether to use it, but how to integrate it into a governed program without adding risk.

40%

SAP-targeted efficiency gain in custom code transformation through agentic tooling in 2026^[4]

50–80%

Reduction in as-is/to-be documentation and fit-gap analysis time, per SAP's own CRO^[13]

85%

Reduction in testing effort reported by Panaya across S/4HANA migrations^[1]

48 hrs

Time to generate a full conversion blueprint and code-remediation plan from a system scan^{[1][6]}

What has changed in 2026

Three things converged. First, SAP shipped the Custom Code Migration Agent inside Joule for Developers, an agentic loop that runs ATC analysis, categorizes findings by complexity, applies high-confidence fixes autonomously, and inserts lower-confidence proposals as commented code for developer review.^{[2][4]} Second, specialist platforms (KTern, Panaya, SNP, CeleRITE) matured to the point where AI-driven readiness analysis, test generation, and code remediation are standard features rather than experiments.^{[1][6][7]} Third, general-purpose GenAI is now reliably applicable to the non-technical labor, requirements documentation, fit-gap write-ups, functional specifications, training materials, and status reports, which absorbs a disproportionate share of consultant hours on any large program.^{[13][16]}

What does not change

AI accelerates; it does not replace governance, judgment, or human review on material decisions. The organizations getting the most from AI acceleration are not the ones deploying the most tools; they are the ones that mapped their bottlenecks first, chose the highest-impact interventions, and kept a human in the loop on every output that goes into a client sign-off or a production system. The risks of uncritical AI use, hallucinated functional specs, untested code fixes, and false test coverage, are real and documented.^[15]

The decisions a program leader must make

1. **Where to intervene.** AI is not uniformly valuable across all phases. Map your bottlenecks and apply AI where the manual effort is highest and the output can be verified (Section 3).
2. **Which tooling layer.** SAP-native (Joule for Developers, Signavio, SAP Cloud ALM), specialist migration platforms, or general-purpose GenAI, each has a different scope, cost, and governance burden (Section 11).
3. **How to govern the output.** Every AI-generated artifact needs a review protocol, confidence threshold, and accountable human owner before it enters the delivery chain (Section 12).

BOTTOM LINE FOR THE PROGRAM SPONSOR

An AI-enabled S/4HANA program running today should target 25 to 40% compression in total delivery effort on the highest-volume, most repetitive workstreams: code scanning, test case generation, documentation drafting, and status reporting. That compression does not reduce the need for experienced consultants; it redirects them from mechanical work to judgment, governance, and business engagement, which is where overruns actually originate.

[13][15]

2

Why AI Changes the Migration Equation

S/4HANA migration has always been expensive and slow for the same reasons: enormous volumes of code to assess, processes to document, and test cases to write, all requiring specialized knowledge that is in short supply. AI attacks exactly those bottlenecks.

2.1 The cost structure AI disrupts

Implementation consulting represents 40 to 60% of total migration program cost.^[14] The largest share of that consulting time is not strategic; it is mechanical: reading ABAP objects to identify deprecated calls, writing functional specifications from workshop notes, creating test scripts from process documentation, and drafting status reports from spreadsheet data. These are exactly the tasks GenAI and agentic tools handle well, at scale, without fatigue.

Before AI: where the time goes

- 4 to 8 weeks of manual workshops to map as-is processes
- Months of ABAP developer effort to scan, classify, and remediate Z-objects
- Test case creation driven by tribal knowledge, often with large gaps
- Documentation written from scratch by senior consultants
- Status reports assembled from disconnected spreadsheets and emails
- Data mapping done manually, object by object

With AI: what changes

- System scan delivers a complete landscape map in 48 hours
- Agentic code migration fixes high-confidence issues in batch; developers review exceptions
- Test cases generated from process models, usage logs, and requirements automatically
- Functional specs, training materials, and release notes drafted in minutes
- Real-time dashboards from a single governance platform, not assembled by hand
- LLM-assisted data mapping with lineage, anomaly detection, and suggested transformations

2.2 The 2026 tooling inflection

Earlier AI tooling for SAP migration was largely diagnostic: scan and report, with human remediation. The 2026 shift is toward agentic: scan, plan, fix, validate, and report, with human oversight by exception. SAP's Custom Code Migration Agent is the clearest example, applying deterministic fixes autonomously and inserting AI-generated proposals for developer review where confidence is lower.^{[2][4]} SAP Cloud ALM has also expanded to serve as a central governance and quality-gate platform for the full Activate lifecycle, including AI-assisted status and risk signals.^[15]

2.3 The consultant role it creates

The consultant who previously spent half their time on documentation and code scanning is now a reviewer, decision-maker, and business bridge. The skills that matter most on an AI-enabled program shift toward judgment

on AI output quality, prompt design for reliable results, exception handling on complex cases, and stakeholder engagement that automation cannot replace. This is also why AI acceleration does not reduce the need for experienced SAP practitioners; it increases the leverage of each one.

WHERE SOURCES DIFFER: VENDOR CLAIMS VERSUS FIELD EXPERIENCE

Vendor-reported figures for AI acceleration (80 to 98% reductions in code remediation effort, for example) are drawn from favorable deployments and should be treated as ceilings, not baselines.^{[1][7]} Independent practitioners and SAPinsider research report more modest but still material gains: 25 to 50% compression in specific workstreams, with the caveat that data quality and program governance are the binding constraints, not the sophistication of the AI tool. Use vendor figures to justify the tooling conversation; use independent benchmarks to set expectations with the steering committee.

3 The AI Acceleration Map: Phase by Phase

AI does not accelerate every phase equally. The highest-value interventions concentrate in Discover, Explore, and Realize, the phases where manual effort is most repetitive and volume is highest. The map below positions AI across the full SAP Activate lifecycle.

FIGURE 3.1 · AI ACCELERATION ACROSS THE SAP ACTIVATE PHASES

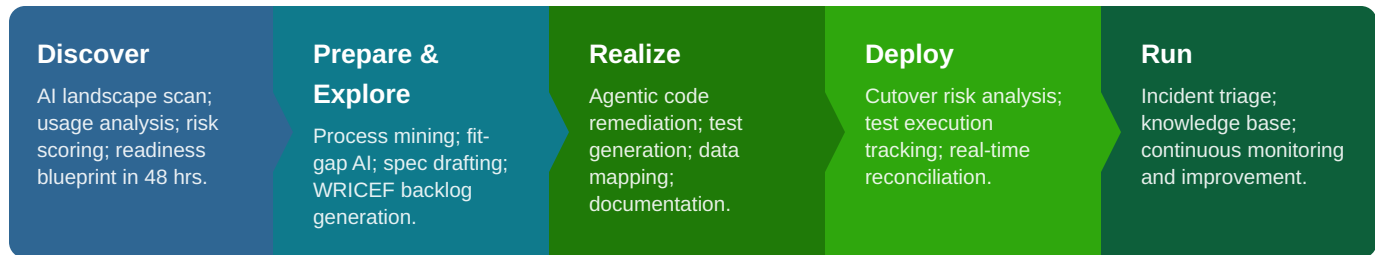
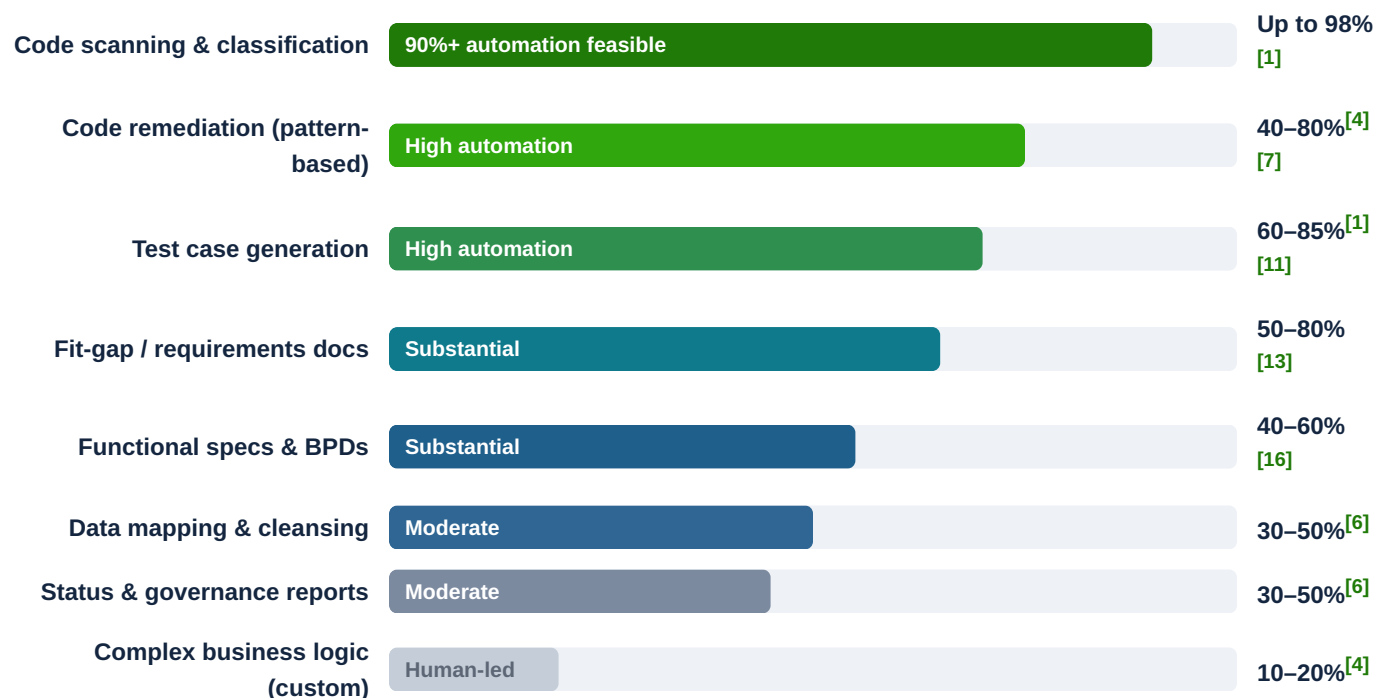


FIGURE 3.2 · INDICATIVE EFFORT REDUCTION BY WORKSTREAM



Ranges synthesized across vendor, SI, and practitioner sources. Upper bounds reflect favorable conditions; plan conservatively and measure against your own baseline.

4

Discovery & Readiness: From Weeks to Days

The Discover phase sets the credibility of every estimate that follows. Organizations that enter it with manual workshops and spreadsheet-driven landscape mapping consistently discover surprises in Realize, where surprises are expensive. AI-driven discovery eliminates most of those surprises before scope is locked.

4.1 What AI-driven discovery produces

A metadata extract from the ECC system, fed into a specialist platform, now delivers within 24 to 48 hours what previously required several weeks of consultants querying tables and interviewing teams.^{[1][6]} The output includes a complete inventory of custom ABAP objects with usage frequency and risk scoring, a map of all integrations and their compatibility exposure, identification of simplification items relevant to the target S/4HANA release, add-on compatibility analysis, Business Partner conversion scope, sizing estimates, and a risk-scored remediation roadmap.

4.2 The tools that generate it

Tool	What it scans	Key AI output
SAP Readiness Check	Custom code, add-ons, simplification items, sizing, recommended Fiori apps. ^[V1]	Risk-scored conversion blueprint; prioritized remediation list.
KTern.AI	Full ECC landscape including usage patterns, integrations, users, process profiles. ^[6]	Digital map with risk scoring and test recommendations; days not weeks.
Panaya S/4 360	ECC system including code, test, and business-process coverage gaps. ^[1]	Conversion blueprint in 48 hours; pre-built project plan for brownfield.
LeanIX / process intelligence	Application portfolio, integrations, and dependencies. ^[V1]	Architecture map and rationalization recommendations.
SAP Signavio	Business processes via process intelligence and mining. ^[11]	As-is process map with conformance analysis and S/4HANA delta identification.

4.3 What this means for the program

A discovery phase that previously consumed four to eight weeks of consultant effort and still produced incomplete results can now be compressed to one to two weeks for the AI-assisted scan plus interpretation, with significantly more completeness. Critically, usage data replaces assumption: instead of teams listing what they think is critical,

the system shows what actually runs, enabling evidence-based scoping decisions and early decommissioning of unused code that would otherwise consume remediation budget.^{[6][7]}

PRACTITIONER PRINCIPLE: SCAN BEFORE YOU SCOPE

The single most reliable use of AI budget on an S/4HANA program is a platform-driven landscape scan before the Statement of Work is finalized. It is the cheapest point to discover that 30% of Z-objects are unused, that a critical integration has no S/4HANA-compatible version, or that data quality is three months of remediation work rather than three weeks. Discoveries made after kick-off cost orders of magnitude more to accommodate.^{[6][13]}

5

Requirements & Fit-Gap: From Workshops to Continuous Analysis

Fit-to-standard workshops are the heart of the Explore phase and one of the most labor-intensive parts of any SAP implementation. GenAI restructures how that work is prepared, captured, and converted into signed-off deliverables.

5.1 Process mining as the starting point

Rather than entering fit-gap workshops with blank templates and participant memory, AI-enabled teams start with a process map generated from actual system usage. SAP Signavio's process intelligence module mines the ECC transaction log to produce a conformance-checked as-is model that shows how processes actually run, including variants and exceptions, not how they were designed to run.^[11] Workshop time then focuses on delta decisions rather than description: the as-is is pre-documented, and the conversation is about what the to-be should look like.

5.2 GenAI for requirements capture and specification

SAP's own Chief Revenue Officer for Cloud ERP cited a 50 to 80% reduction in the time needed to document current processes and assess the gap between as-is and to-be using GenAI.^[13] In practice this surfaces in several specific workstreams:

- **Workshop transcription and synthesis.** GenAI converts raw workshop notes or recordings into structured fit-gap logs, business process descriptions, and open-item lists, in a fraction of the time a consultant would take.
- **Functional specification drafting.** Given a process description and a gap statement, GenAI drafts a structured functional specification that a business analyst then reviews and refines rather than authors from scratch.
- **WRICEF backlog generation.** AI tools can analyze gap logs to identify and categorize the custom development objects (Workflows, Reports, Interfaces, Conversions, Enhancements, Forms) that follow, populating the WRICEF register automatically from workshop outputs.^[9]
- **User story conversion.** Requirements captured as free text or workshop outputs can be converted to structured user stories or Activate-format process requirements at scale.

FIGURE 5.1 · EXAMPLE GENAI PROMPT PATTERNS FOR FIT-GAP WORK

WORKSHOP SYNTHESIS	FUNCTIONAL SPEC DRAFTING	WRICEF CLASSIFICATION
"Given these workshop notes [paste], identify: (1) processes that fit SAP standard, (2) gaps requiring custom development, (3) open questions needing business decision. Format as a fit-gap	"Draft a functional specification for the following gap: [describe gap]. Use this template: [paste section headings]. Include: business requirement, current process, proposed S/4HANA solution, data	"Review this list of gaps [paste]. For each, classify as: W (Workflow), R (Report), I (Interface), C (Conversion), E (Enhancement), or F (Form). Add an estimated development effort (S/M/L) and

register with a column for complexity (low/medium/high) and proposed resolution."

fields affected, integration impact, and open questions."

whether a standard SAP alternative exists in S/4HANA."

THE REVIEW OBLIGATION

Every AI-generated functional specification must be reviewed and signed off by a qualified SAP functional consultant and a business process owner before it enters the development backlog. GenAI drafts are starting points, not deliverables. The error risk is not that the AI misses the point; it is that it produces a plausible-looking but incorrect specification that passes a casual review. Build the review step into the workflow as a non-negotiable gate.^[15]

6

Custom Code Remediation: The Biggest Bottleneck, Now Automated

Custom ABAP code remediation is consistently the single largest source of delay in S/4HANA migrations. The average enterprise carries more than 2,000 custom objects; the analysis, classification, remediation, and testing of these objects has historically consumed the majority of technical delivery effort. Agentic AI is now the most material accelerator available for this workstream.

6.1 The SAP Custom Code Migration Agent

Released in 2026 within Joule for Developers, the SAP Custom Code Migration Agent is the most significant native tooling development for migration programs in years.^[2] It operates as an autonomous agentic loop within ABAP Development Tools (ADT), running the full remediation sequence with human oversight only where needed.

Agent step	What it does	Human role
ATC analysis	Runs the ABAP Test Cockpit across all custom code packages in batch mode, surfacing all S/4HANA compatibility findings. ^[2]	Initiates; monitors progress.
Intelligent interpretation	Categorizes findings by complexity and available remediation options (deterministic quick fix, AI-generated fix, or manual). ^[2]	Reviews category assignments.
Deterministic fixes	Applies documented, rule-based fixes (deprecated SELECT *, obsolete function modules, compatibility view replacements) automatically at scale. ^[4]	Batch approval or sampling review.
AI-generated fixes	For each AI fix, calculates a confidence score. High-confidence fixes are applied; lower-confidence fixes are inserted as commented code proposals for developer review. ^[2]	Reviews and accepts or modifies proposals.
Unit test generation	After each fix, generates ABAP unit tests to validate the modified logic produces the same outputs as the original code. ^[4]	Reviews test coverage; approves.
ATC re-validation	Re-runs ATC on fixed objects to confirm all flagged issues are resolved before marking the object migration-ready. ^[4]	Reviews summary report.

SAP has confirmed a target of 40% efficiency gain in custom code transformation through this agentic tooling, meaning a remediation program that previously required 18 months of ABAP developer effort could be completed in 10 to 11 months at the same headcount, or faster with additional developer capacity applied to the human review steps.^[4]

6.2 Specialist third-party platforms

Several specialist platforms offer comparable or complementary capabilities, often with deeper integration into the broader migration program context.

Platform	Key AI capability	Reported outcome
Panaya (Seemore AI)	AI assistant analyzes business logic and code context to suggest safe corrections; developer reviews in natural language; integrates with SAP transport process. ^[5]	Months to two to four weeks for complex remediation; up to 98% reduction in remediation effort. ^{[1][5]}
Appexus CeleRITE	Automates up to 70 to 80% of custom code remediation; generates documentation at RICEFW level; scans for security vulnerabilities. ^[7]	Vendor-reported: most code remediation automated; documentation generated automatically. ^[7]
KTern.AI	AI-driven code impact analysis with risk scoring; automated test recommendations from usage data. ^[6]	Discovery that normally takes months done in days; often 40%+ faster overall timelines. ^[6]
Cognizant SmartMove	Gen-AI based brownfield conversion covering data validation, reconciliation, and code remediation. ^[8]	SAP-qualified; ISG-recognized leader for RISE with SAP implementation. ^[8]

6.3 What AI handles well versus what it does not

A critical distinction for program planning: AI accelerates pattern-based remediation. It is less reliable on complex business logic.

AI handles well (high automation feasible)	Requires human expertise (AI assists only)
Deprecated SELECT * and obsolete function module replacements. ^[4]	Complex proprietary business logic embedded in Z-programs.
Compatibility view replacements for removed database tables.	Custom code that calls external systems or has undocumented dependencies.
Syntax and standard-compliance fixes across large volumes.	Enhancements that need redesign rather than remediation.

AI handles well (high automation feasible)	Requires human expertise (AI assists only)
Unit test generation for modified objects. ^[4]	Business Partner / CVI conversion with complex vendor data.
Dead-code detection and decommissioning recommendations.	Authorization and security object redesign for Fiori roles.

THE CLEAN-CORE CONNECTION

Code remediation is the moment to enforce clean-core discipline. The question for every Z-object is not only "can we migrate it?" but "should we?" AI usage data analysis makes the decommissioning case objectively: objects with zero or near-zero usage in the past two years should be retired, not remediated. Every object retired before migration is one that will not accumulate technical debt in S/4HANA and block future AI capabilities gated behind clean-core compliance.^{[4][14]}

7

Testing: From Spreadsheet-Driven UAT to AI-Generated Coverage

Testing is the phase most likely to compress or expand based on preparation quality, and also the one where AI has the most mature and widely deployed tooling. Organizations still running SAP testing from spreadsheets and manual scripts are leaving the largest available acceleration on the table.

7.1 The testing problem at scale

A full S/4HANA migration requires coverage across every modified business process, every integration, every WRICEF object, and every Fiori role, with multiple test cycles: unit, string, integration, regression, and user acceptance. At the volume of a large brownfield program, this can mean thousands of test cases, each requiring definition, data, execution, and documentation. Traditionally this work is manual, driven by consultants with tribal knowledge of the business processes, and produces incomplete coverage because the knowledge is never fully captured.^[11]

7.2 How AI transforms each testing stage

Testing stage	AI capability	Tool examples
Test case generation	GenAI analyzes requirements documents, Signavio process models, user stories, and system configurations to generate comprehensive, structured test cases, including edge cases often missed in manual design. ^[11] ^[12]	SAP Signavio + GenAI; ImpactQA; Tricentis; Worksoft.
Test data creation	AI generates realistic, variant-rich test data from actual usage patterns and master data profiles, avoiding the common bottleneck of teams waiting for test data to be manually prepared. ^[12]	Specialist data generation platforms; KTern.AI.
Test script generation	Test cases written in natural language are automatically translated into executable scripts for automation frameworks (Python, RPA, Selenium), eliminating manual scripting effort. ^[10] ^[12]	Tricentis Tosca AI; Worksoft Certify; ImpactQA.
Risk-based test prioritization	AI analyzes code changes, business process impact, and historical defect patterns to identify which test cases are most critical to execute, optimizing coverage within time-constrained windows. ^[12]	Panaya; KTern; ImpactQA.
Defect analysis and prediction		AI-enabled test management platforms.

Testing stage	AI capability	Tool examples
	AI identifies patterns in execution results to predict defect hotspots, suggest root causes, and prioritize bug fixes, reducing resolution time. ^[12]	
Regression automation	End-to-end regression suites maintained by AI to flag which tests are affected by each code change, so regression scope is always right-sized and never guessed. ^[1]	Panaya Test Management; Tricentis.

7.3 The Signavio-to-test pipeline

One of the most efficient patterns available in 2026 is the direct pipeline from SAP Signavio to AI-generated test cases. Signavio produces business process models in BPMN 2.0, and GenAI can analyze those process models to generate detailed test cases across SAP GUI or Fiori screens, including step-level instructions, expected results, and test data requirements, in minutes rather than weeks.^{[11][12]} These outputs can then be converted into automation scripts for the test execution framework of choice. The practical effect: a business process documented in Signavio becomes a tested and automated scenario within the same week rather than the same sprint.

COVERAGE, NOT JUST SPEED

The underrated benefit of AI-generated testing is coverage breadth, not just speed. Human-designed test cases are constrained by what the designer knows and remembers; AI-generated cases from process models and requirements systematically cover the full documented scope including variants and edge cases that manual design routinely misses. Around 65% of S/4HANA programs report serious quality issues after go-live;^[V1] AI-expanded test coverage is the most direct technical intervention available to reduce that number.

8

Data Migration: Cleansing, Mapping, and Validation at Scale

Data quality issues are the single largest source of budget overrun in S/4HANA programs, with programs routinely budgeting 10 to 15% of total project cost for data work and still discovering surprises late.^[V1] AI cannot fix bad data, but it can find it faster, map it more accurately, and validate it more completely than manual methods allow.

8.1 AI-assisted data cleansing

In the legacy system, years of inconsistent master data accumulate: duplicate vendors, mismatched customer records, orphaned cost centers, and unreconciled open items. AI tools can scan master data at scale, identify duplicates through fuzzy matching rather than exact-match rules, flag referential integrity violations, and surface anomalies that simple validation rules would miss. For vendor data specifically, which is the classic and expensive failure mode in S/4HANA migrations, AI-driven deduplication can process hundreds of thousands of records and surface consolidated candidates for human review, compressing weeks of manual work to days.

8.2 LLM-assisted data mapping

Data mapping, translating legacy field structures to S/4HANA target structures, is one of the most knowledge-intensive manual tasks in a migration. LLMs trained or prompted with source and target data dictionaries can propose field mappings, identify likely transformations, and flag fields with no obvious target, producing a first-draft mapping that a data architect reviews and refines rather than builds from scratch. For organizations with non-SAP legacy systems feeding into S/4HANA, this capability is particularly significant, as the mapping complexity is highest where tool support has historically been weakest.^[9]

8.3 AI validation and reconciliation

Post-load validation, confirming that what arrived in the target system matches what was extracted from the source, is traditionally manual and time-consuming. AI-enabled reconciliation tools can compare source and target record counts, aggregate values, and referential integrity at scale in hours rather than days, with anomaly detection that flags statistical outliers for human review rather than requiring line-by-line comparison.

Data task	AI acceleration	What remains human
Duplicate detection	Fuzzy-match AI across full master data volumes; priority-ranked merge candidates. ^[9]	Merge decision on ambiguous cases; business rule definition.
Field mapping	LLM proposes source-to-target mapping with confidence scores from data dictionary context.	Architect review; complex transformations; business rule validation.

Data task	AI acceleration	What remains human
Data quality scoring	Automated profiling of completeness, consistency, and validity at field and record level.	Root-cause analysis; remediation prioritization decisions.
Archiving scope	Usage analysis identifies data below the active retention threshold; archiving candidates flagged automatically.	Business sign-off on retention policy; legal and compliance review.
Post-load reconciliation	Statistical comparison of source versus target; anomaly detection and exception flagging.	Anomaly investigation; business-critical balance confirmation.

9

Documentation & Knowledge Transfer: From Labor-Intensive to Automated

Documentation is the workstream most universally under-resourced on SAP programs, not because it is unimportant but because experienced consultants are pulled to delivery tasks and documentation falls behind. GenAI makes it possible to produce documentation in parallel with delivery rather than in arrears.

9.1 Where GenAI creates the most value

Document type	AI approach	Review requirement
Business Process Documents (BPDs)	Generated from Signavio process models or workshop notes; structured to SAP Activate template. ^[16]	Functional consultant and process owner review.
Functional specifications (FSDs)	Drafted from gap statements and WRICEF entries; first-pass in minutes rather than days. ^[16]	Senior consultant review; technical peer review.
Remediated code documentation	AI generates inline comments and change logs for every object touched during remediation; reduces future maintenance risk. ^[7]	Sampling review by ABAP lead.
Training materials	Role-based training content drafted from process documentation and system configuration; converted to eLearning formats. ^[16]	Business process owner validation; training lead review.
Release notes	Auto-generated from transport logs, change request descriptions, and configuration records at each deployment cycle.	Change manager review and approval.
Knowledge base articles	Incident patterns from hypercare converted to searchable knowledge articles automatically; reduces recurring support load. ^[7]	Support lead validation before publishing.

9.2 The hidden cost this addresses

Documentation deficits compound after go-live. A system that goes live without complete process documentation, up-to-date technical specs, and role-based training materials creates support costs, user adoption problems, and upgrade risk that are far more expensive than the documentation work itself. The cost of AI-assisted documentation on a large program is marginal; the cost of the documentation gaps it closes is not.

FIGURE 9.1 · EXAMPLE GENAI PROMPT PATTERNS FOR DOCUMENTATION

BUSINESS PROCESS DOCUMENT

"Based on this process description and S/4HANA configuration [paste], draft a Business Process Document with sections: process overview, trigger, steps (numbered), roles (RACI), system transactions (T-codes / Fiori apps), integration points, exceptions and escalations, and key controls."

TRAINING MATERIAL

"Create a role-based quick-reference guide for [role name] performing [process name] in S/4HANA Fiori. Include: when to use this process, step-by-step instructions with screen names, common errors and how to resolve them, and who to contact for help."

RELEASE NOTES

"Based on these transport descriptions [paste], draft concise release notes for [audience: business users / IT / steering]. Group by business area. For each change: what changed, why it changed, what users need to do differently, if anything."

Project Governance & Reporting: Real-Time, Not Retrospective

Program governance is where overruns originate and where they are caught late. AI-enabled governance platforms shift the signal from retrospective (last week's status from spreadsheets) to real-time (continuous monitoring of delivery against plan, risk flags, and quality gates).

10.1 What AI-enabled governance looks like

Platforms such as KTern and SAP Cloud ALM, when used as the program's central delivery system rather than just a repository, provide automated tracking of workstream progress, test execution rates, defect trends, and critical-path risk flags. AI can analyze patterns across these signals to surface early warnings: a testing workstream running behind plan, a module with a rising defect rate that signals a design problem, or a data migration stream where cleansing velocity is insufficient to meet the cutover window.^{[6][15]}

10.2 GenAI for steering committee reporting

Every program generates the same reporting burden: weekly status reports, risk logs, steering committee packs, and milestone summaries. These are high-visibility documents that consume several hours of program management time per week and yet are largely formulaic. GenAI, given current data from the project management platform, can draft these documents in minutes, with the program manager's role reduced to validation and judgment calls rather than data assembly and formatting.

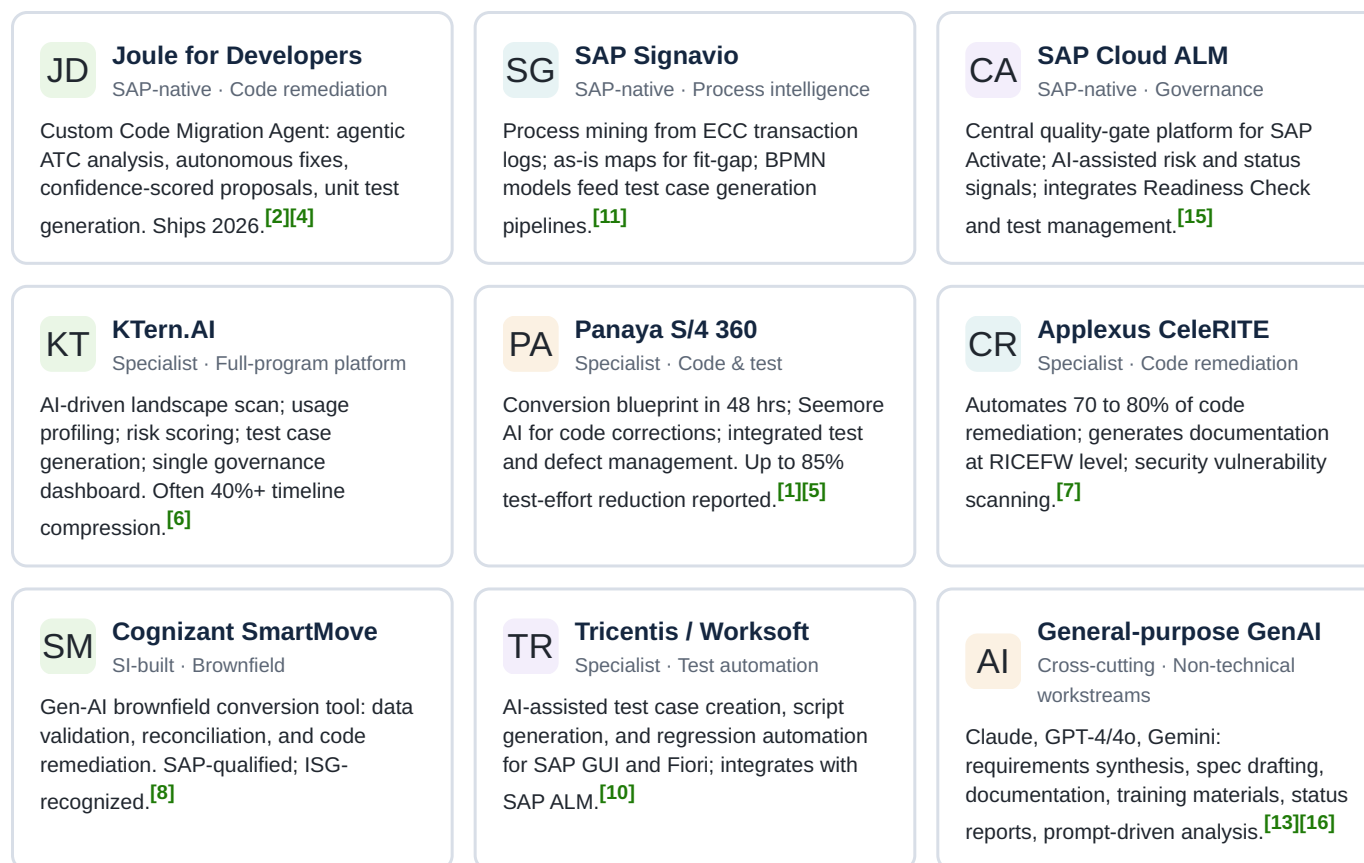
THE ONE-PLATFORM PRINCIPLE

The governance benefit of AI tooling is greatest when the program runs on a single platform, not when status data is collected from multiple disconnected sources and assembled manually before being fed to AI. KTern's value, for example, is contingent on it being the actual delivery system for workstreams, deliverables, sign-offs, and test tracking, not a parallel reporting layer.^[6] Choose your governance platform before kick-off and make it the system of record from day one.

11 The AI Tooling Ecosystem

The tooling landscape has three layers: SAP-native capabilities embedded in the platform, specialist migration acceleration platforms, and general-purpose GenAI applicable to the non-technical workstreams. A well-configured program uses all three in complementary roles.

FIGURE 11.1 · THE AI TOOLING ECOSYSTEM FOR S/4HANA MIGRATION



VENDOR-NEUTRALITY NOTE

The tools listed above are referenced because they are documented in published sources current to mid-2026 and represent the major categories of available tooling. This playbook does not endorse any specific vendor. Program teams should evaluate tools against their specific landscape (SAP release, approach, cloud or on-premise), their SI partner's tooling commitments, and their own governance and security requirements before committing.

12 Risks, Limits & Responsible Use

AI acceleration introduces new failure modes alongside the benefits it delivers. These risks are manageable with the right controls, but programs that deploy AI without a governance framework for its outputs will encounter them.

Risk	How it surfaces	Mitigation
Hallucinated specifications	AI drafts plausible but incorrect functional specs that pass casual review and enter the development backlog. ^[15]	Mandatory sign-off by qualified functional consultant and process owner on every AI-generated spec; never treat as final without review.
Untested code fixes	AI-generated code remediation applied in batch without sufficient unit and integration testing; defects surface in UAT or production. ^[4]	Confidence thresholds, AI-generated unit tests, ATC re-validation, and sampling review as non-negotiable gates in the remediation workflow.
False test coverage	AI-generated test cases cover the documented scope but miss undocumented business variants; go-live defects on processes that appeared tested.	Supplement AI-generated cases with usage-data-driven scenario extraction; validate coverage with business process owners, not just testers.
Vendor figure inflation	Program committed to stakeholders on vendor-reported acceleration numbers (up to 98% reduction) that reflect favorable conditions, not your landscape.	Set internal targets based on independent benchmarks (25 to 40% compression); use vendor figures to justify tooling investment, not to set delivery dates.
Data privacy in AI prompts	Consultants paste client data, including sensitive financial or personal data, into public AI services. ^[15]	Define and enforce a clear policy on what data can be submitted to which AI service; anonymize or use on-premise models for sensitive data workstreams.
Skill gap in AI governance	Program team knows how to use AI tools but not how to design review workflows, confidence thresholds, or exception-handling protocols.	Designate an AI delivery lead responsible for governance protocols; include AI output quality in QA reviews and quality gates.
Over-automation of complex logic	AI applied to custom code with deep proprietary business logic; fixes are syntactically correct but semantically wrong.	Use complexity scoring to route AI for pattern-based remediation and human developers for business-logic-intensive objects; never skip integration testing.

13

Getting Started: A 30-Day AI Acceleration Baseline

The highest-value action in the first thirty days is establishing an evidence base, not deploying tools. The evidence base tells you where AI acceleration has the most impact in your specific landscape, which tools fit your governance constraints, and what the realistic compression targets are.

13.1 The first five moves

1. **Run an AI-driven landscape scan.** Use the SAP Readiness Check and one specialist platform (KTern, Panaya, or equivalent) to generate a system-level view of your custom code volume, usage patterns, integration exposure, and data quality signals. This is the foundation for every subsequent decision.
2. **Map your bottlenecks.** Using the landscape data, identify which workstreams in your specific program have the highest manual effort and the greatest compression potential. Prioritize AI investment there, not uniformly across all phases.
3. **Define your governance protocol.** Before any AI output enters a delivery artifact, define the review workflow: who reviews, what the acceptance criteria are, and how AI-generated content is tracked and versioned. Document this in the project's quality-management framework.
4. **Set conservative targets.** Align with the steering committee on realistic compression expectations based on independent benchmarks rather than vendor ceiling figures. Build in a measurement baseline so actual acceleration is tracked and visible.
5. **Start with code and testing.** The highest-maturity, best-governed AI tooling is in code remediation and test generation. These are the right places to prove the model before expanding to less-mature use cases like data mapping or requirements drafting.

HOW LABWYZE CAN HELP

labwyze supports program teams and transformation leaders in designing an AI acceleration strategy for their specific S/4HANA program: identifying the highest-impact interventions for their landscape, selecting and evaluating tools without vendor bias, designing governance protocols for AI-generated artifacts, and setting realistic targets that can be defended to a steering committee. The goal is a thirty-day baseline that gives you the evidence to commit to an AI-accelerated delivery model, or to know clearly where it does not apply.

A

Appendix: Glossary, Prompt Patterns & Sources

A.1 Glossary

ABAP Test Cockpit (ATC)

SAP's static analysis tool for checking custom ABAP code against S/4HANA readiness criteria; the starting point for any code remediation program.

Agentic AI

AI systems that perceive context, plan multi-step actions, and execute autonomously within defined guardrails; distinguished from GenAI copilots that produce output for human action.

Clean core

SAP's architectural principle of keeping the digital core close to standard and moving extensions to SAP BTP; required for access to the most advanced Joule capabilities.

Custom Code Migration Agent

SAP's agentic tooling within Joule for Developers, released 2026, that automates the ATC-analyze-remediate-test cycle for ABAP objects.

Deterministic fix

A code correction with a single documented replacement (for example, deprecated function module X maps to standard function module Y); safe to apply automatically at scale.

Fit-to-standard

The SAP Activate discipline of running processes against SAP Best Practices and documenting deviations rather than designing to legacy requirements; the point where scope is won or lost.

GenAI (Generative AI)

AI that generates text, code, or other content from a prompt; applicable to documentation, specification drafting, report generation, and prompt-driven analysis.

Process mining

Automated analysis of event logs (typically from the ECC transaction log) to generate as-is process maps showing how processes actually run, including variants.

RICEFW / WRICEF

The catalog of custom development objects required to close fit-gaps: Workflows, Reports, Interfaces, Conversions, Enhancements, Forms. Managing this backlog is a core Realize-phase governance task.

SAP Signavio

SAP's process intelligence and mining platform; produces BPMN process models from system data that feed AI-driven test case generation and fit-gap analysis.

Z-object

A customer-built ABAP development object (programs, function modules, classes) with names beginning with Z or Y; the custom code estate that must be analyzed and remediated during migration.

A.2 Sources

Sources numbered as cited inline. V1 references the Labwyze SAP S/4HANA Migration Playbook (Volume 1, June 2026), which contains the primary references for migration approach, deployment models, and cost benchmarks.

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5. Panaya, "Automatic Code Corrections" (2026). Seemore AI assistant; natural language interaction; SAP Notes integration. panaya.com
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8. Cognizant SmartMove, "S/4HANA Automation Solution" (2026). Gen-AI brownfield conversion; ISG leadership recognition. cognizant.com
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12. SAPinsider / ImpactQA, "The Transformative Role of GenAI in QA and Testing" (2025). Test case generation; test data creation; risk-based prioritization; defect prediction. sapinsider.org
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14. SoftwareSeni, "Do You Need to Migrate to SAP S/4HANA to Get AI Value?" (2026). Clean-core gating of Joule AI; Forrester assessment. softwareseni.com
15. Truspeq, "SAP S/4HANA Migration Roadmap 2026" (2026). AI-driven roadmap; SAP Cloud ALM governance. truspeq.com
16. SAP Community (GenAI in S/4HANA projects), "How can GenAI be effectively used?" (2026). Requirements analysis; BPD drafting; training materials. community.sap.com

A NOTE ON EVIDENCE QUALITY

Code remediation and testing figures from specialist vendors (Panaya, CeleRITE, KTern) reflect favorable deployments and should be treated as upper bounds. Independent estimates and SAP's own published targets tend to be more conservative (25 to 40% program-level compression). Use the vendor figures to make the case for tooling investment; use conservative independent estimates for planning and steering committee commitments. SAP's Custom Code Migration Agent claims, sourced from SAP Community and partner analysis, are recent (June 2026) and should be confirmed against current product documentation before program commitments are made.